

Forensic Investigation Report – Malicious Chrome Extension & Cryptocurrency Mining Activity

Objective

To investigate suspicious activity reported on a client's business Facebook account, identify any malicious browser extensions or unauthorized activity, determine the scope of compromise, and provide actionable findings for remediation.

Executive Summary

A client reported unusual behavior on their business Facebook account, prompting a full digital forensics review. Initial interviews revealed that an employee had recently installed a browser extension marketed as an AI assistant. Examination of the affected system, using a simulated **Blue Team Labs Online** "Browser Forensics: Crypto Miner" scenario, uncovered a malicious Google Chrome extension.

The investigation centered on an **AD1 logical evidence image** of the browser's profile data, acquired with FTK Imager. Through targeted analysis with FTK Imager and Browser History Viewer, two Chrome profiles were identified, one of which contained the suspicious extension. Timeline correlation between browsing history and extension installation confirmed the presence of a benign theme ("Earth in Space") alongside a malicious extension named "**DFP Cryptocurrency Miner**."

Code review of the extension revealed embedded JavaScript linked to **CryptoLoot**, a browser-based cryptocurrency mining framework. The miner operated at 20 hashes per second, using a unique public key to direct earnings to the attacker's account. The extension was capable of running in the background, consuming system resources, and potentially enabling additional malicious behavior.

These findings highlight the security risks posed by browser extensions from unverified sources. The malicious extension was confirmed as the primary threat vector for the reported Facebook account anomalies.

Understanding the Evidence Format

An **AD1 file** is a proprietary "logical image" format created by FTK Imager. Logical images differ from physical disk images because they capture only selected file systems or folders, not the entire raw disk. This makes them useful for targeted investigations, such as browser artifacts, without the overhead of imaging the full drive. Since AD1 is proprietary, specialized forensic tools like FTK Imager are required to parse it.

FTK Imager itself is a forensic acquisition and preview tool. It allows investigators to mount, browse, and extract files from a variety of evidence formats, while maintaining chain-of-custody integrity. It's not a full analysis suite, but is ideal for mounting evidence and previewing its contents.

Preparation of Tools

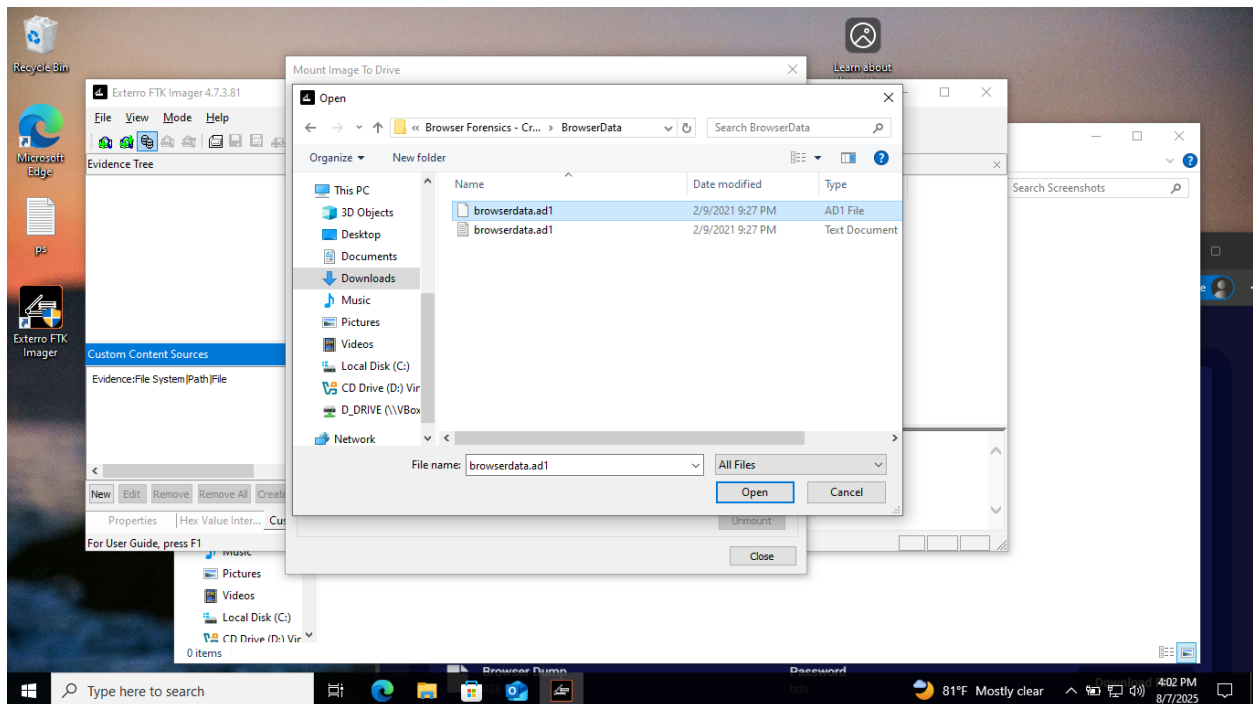
Two main tools were required:

1. **FTK Imager** – for mounting and browsing the AD1 file contents.
2. **Browser History Viewer** – a utility for parsing browser artifacts such as history databases, cache, and profile data into human-readable timelines.

Both tools were downloaded and installed on a 64-bit Windows analysis workstation.

Evidence Acquisition & Extraction

The lab package was downloaded and extracted using 7-Zip. The ZIP archive required a password ("btlo") provided in the lab instructions. Within the extracted directory, a subfolder named `browser_data` contained the `browserdata.ad1` file — the primary artifact for analysis.



Mounting the AD1 in FTK Imager

FTK Imager was launched, and the following steps were performed:

- **File** → **Add Evidence Item** → **Image File** was selected.
 - Navigated to and selected `browserdata.ad1`.
 - The evidence loaded successfully, revealing a top-level folder structure similar to the `C:\Users\` directory on Windows systems.
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Navigating to Browser Profile Data

In a Windows environment, **user-specific application data** is stored in two primary subdirectories under the `AppData` folder:

- **Local** – data stored on the machine only, not intended for roaming between systems.
- **Roaming** – data synchronized with a roaming user profile in a domain environment.

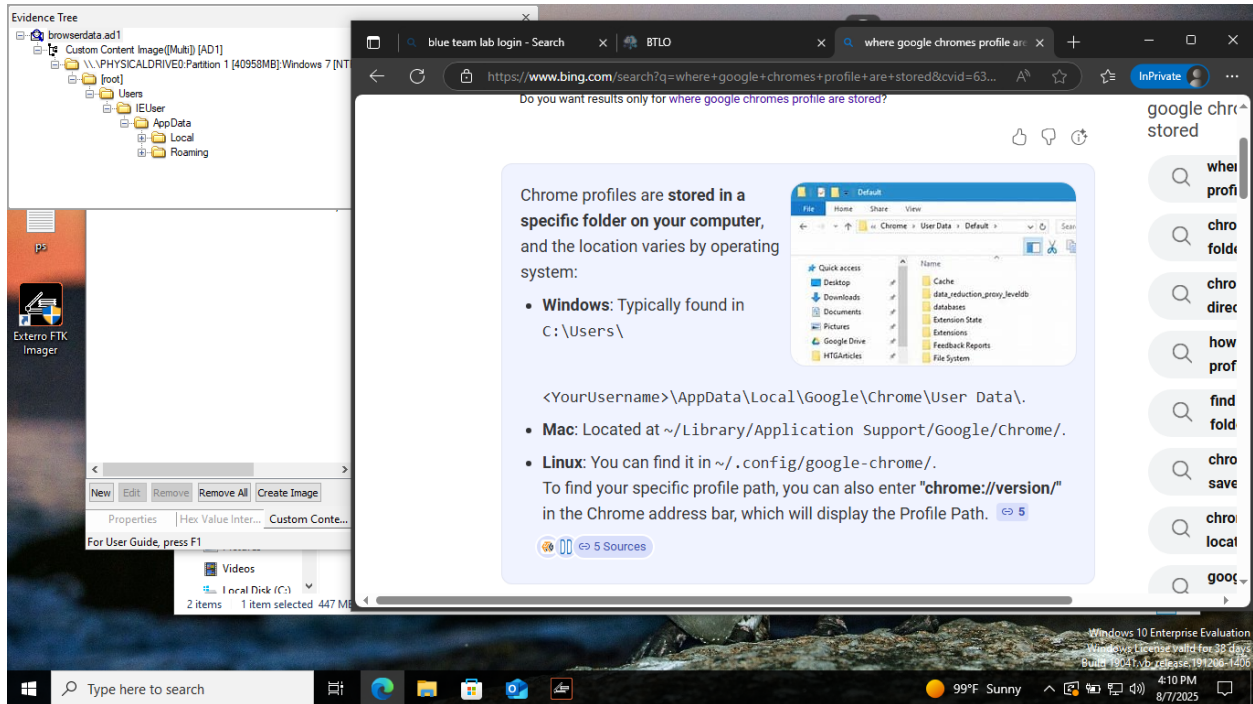
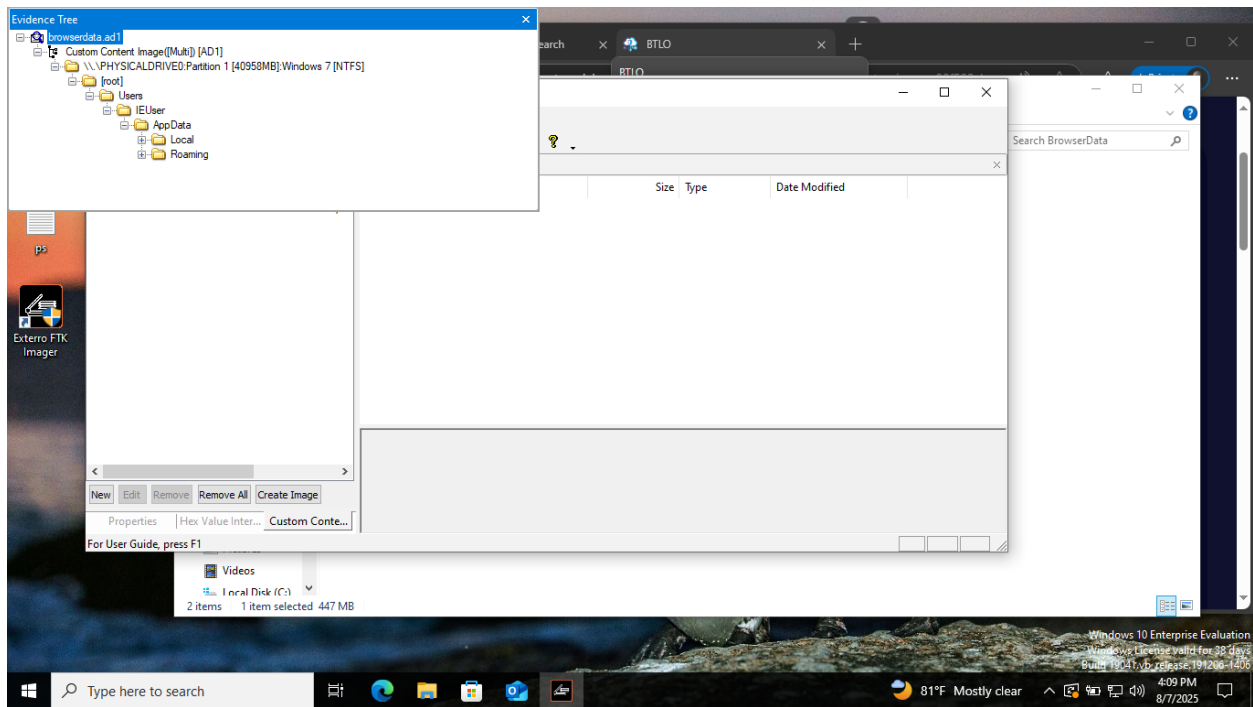
Google Chrome stores profiles under `AppData\Local\Google\Chrome\User Data\`. Each profile is represented as a folder, such as `Default` (primary profile) or `Profile 1`, `Profile 2`, etc.

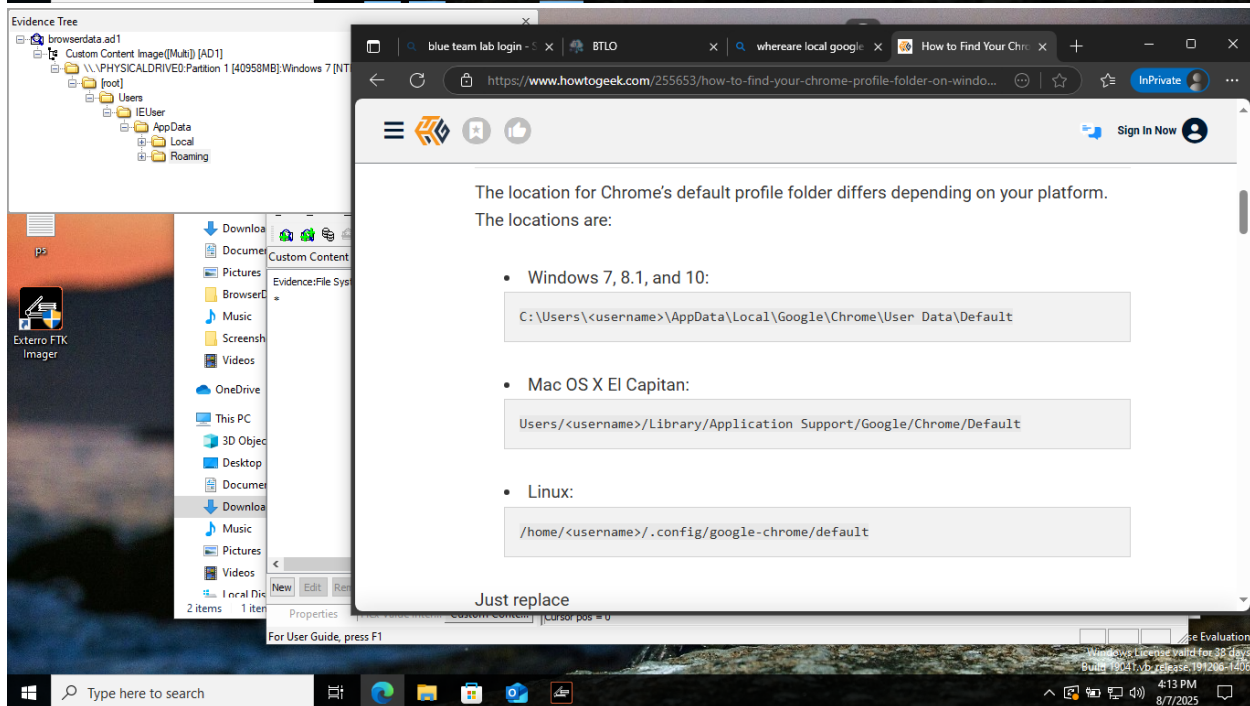
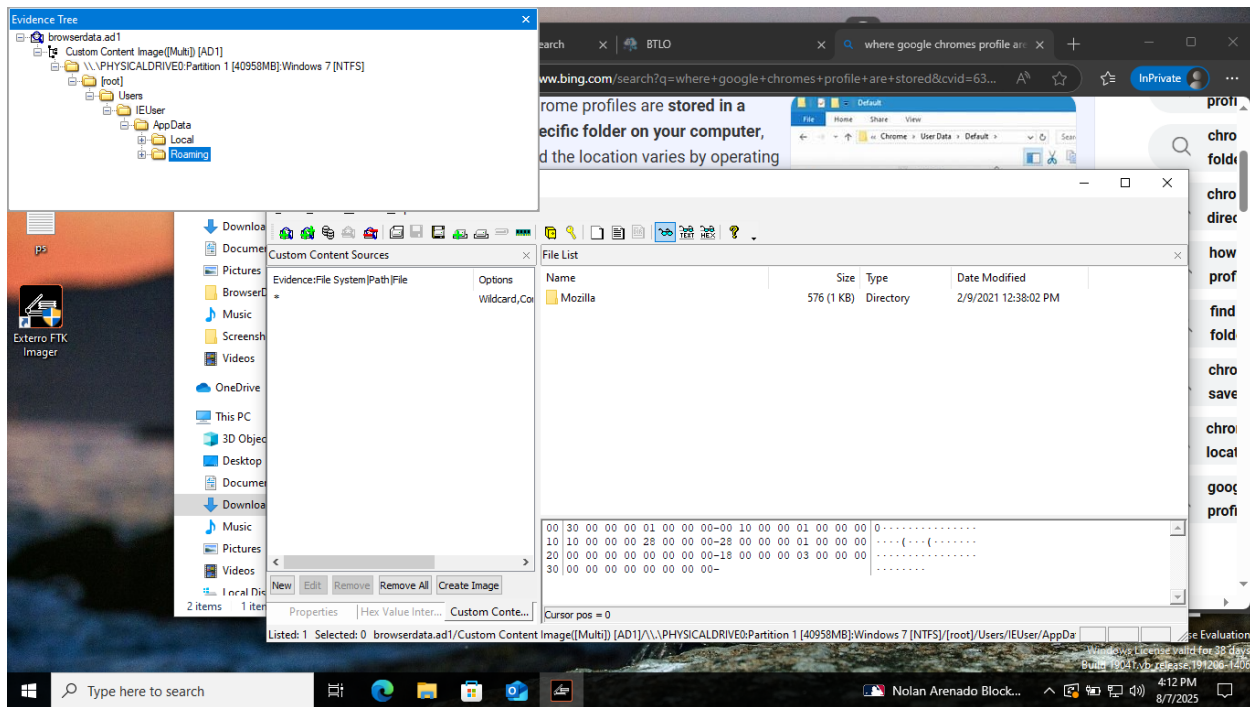
Within the AD1, the path:

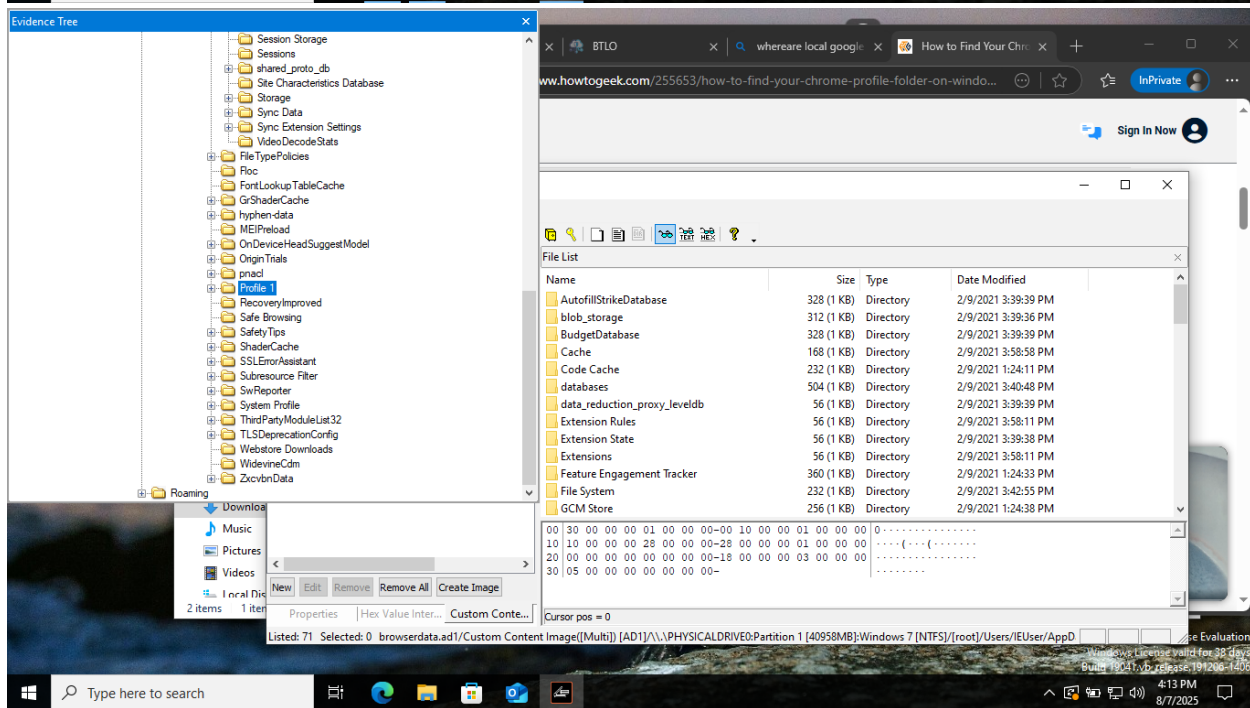
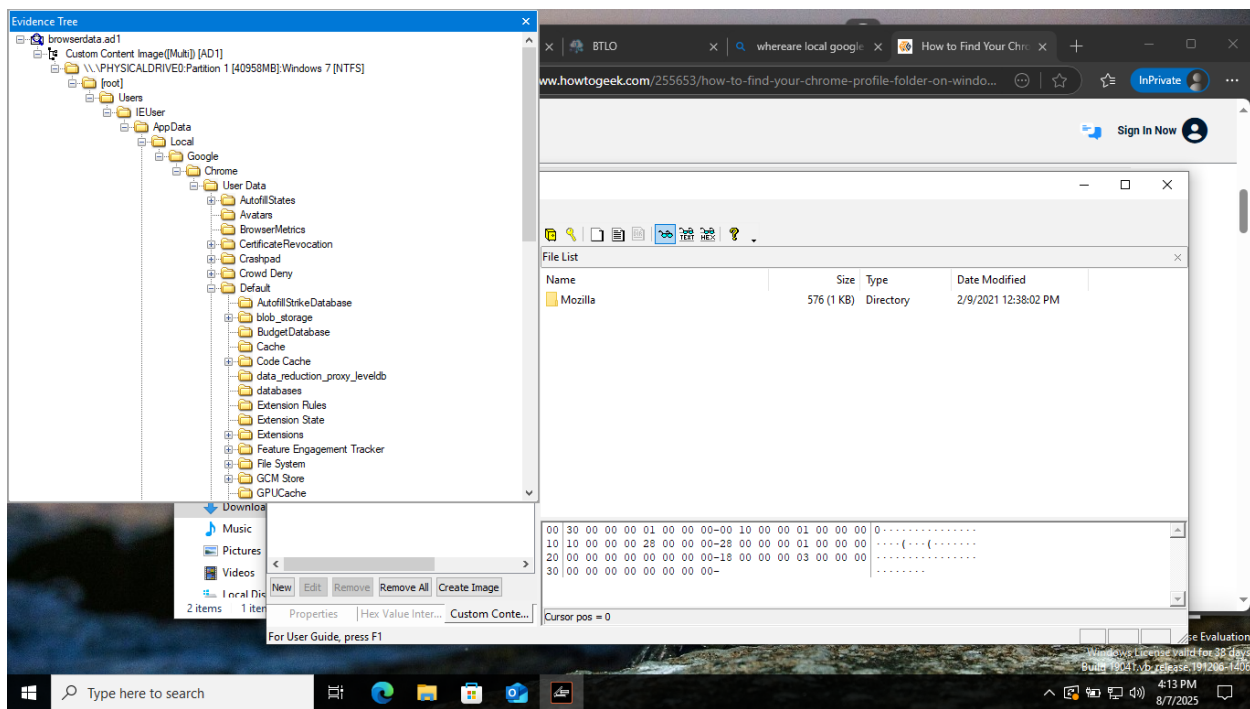
`Users\IEUser\AppData\Local\Google\Chrome\User Data\` contained two profile folders:

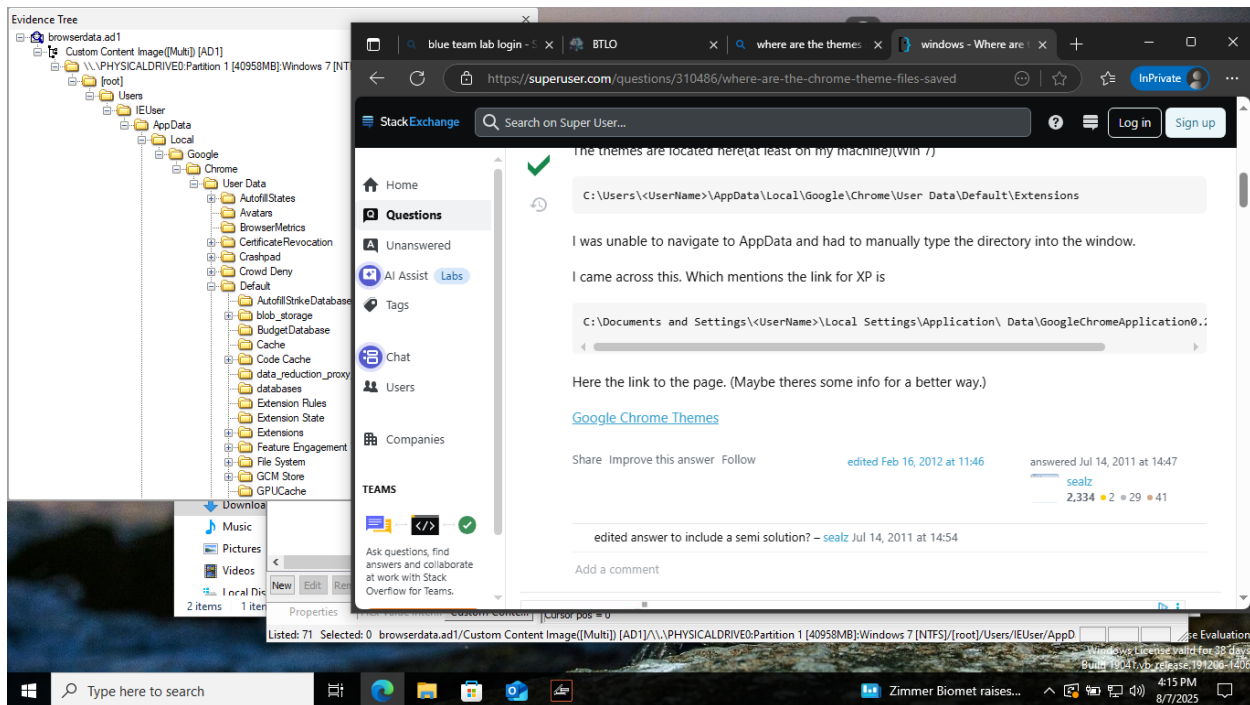
- `Default` – the initial Chrome profile.
- `Profile 1` – an additional user-created profile.

This confirmed that **two distinct Chrome browser profiles** were present.









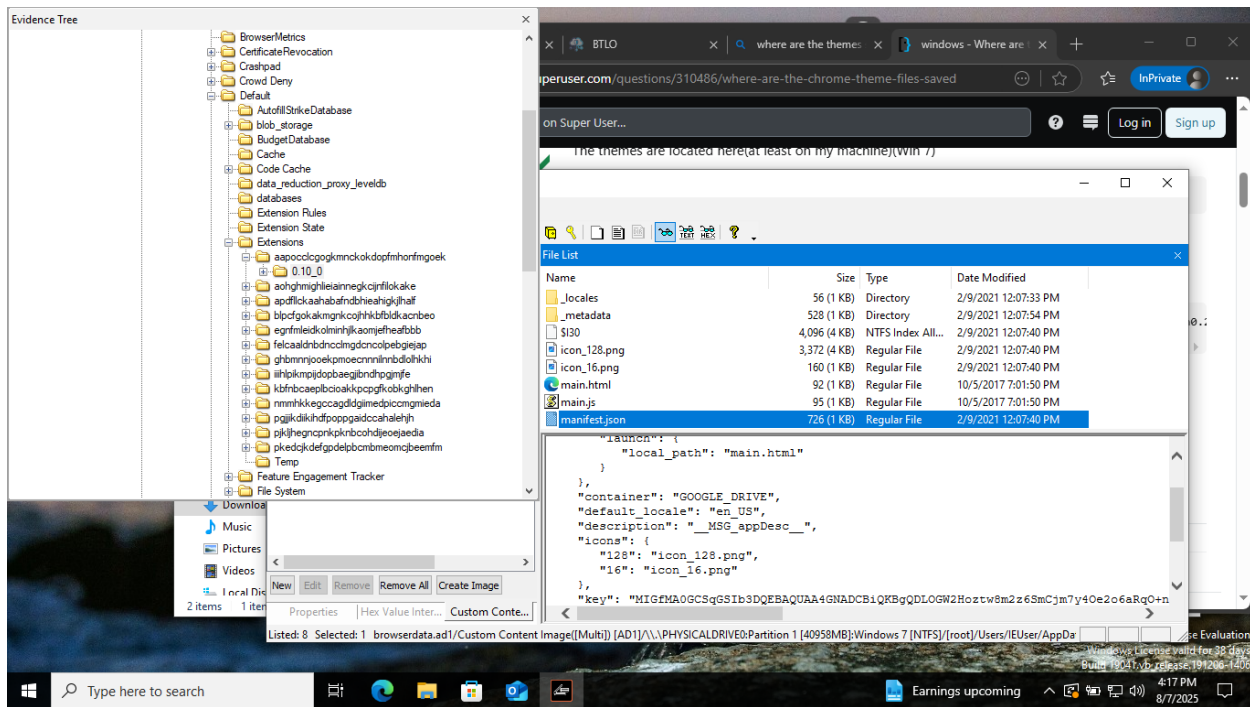
Locating Installed Themes and Extensions

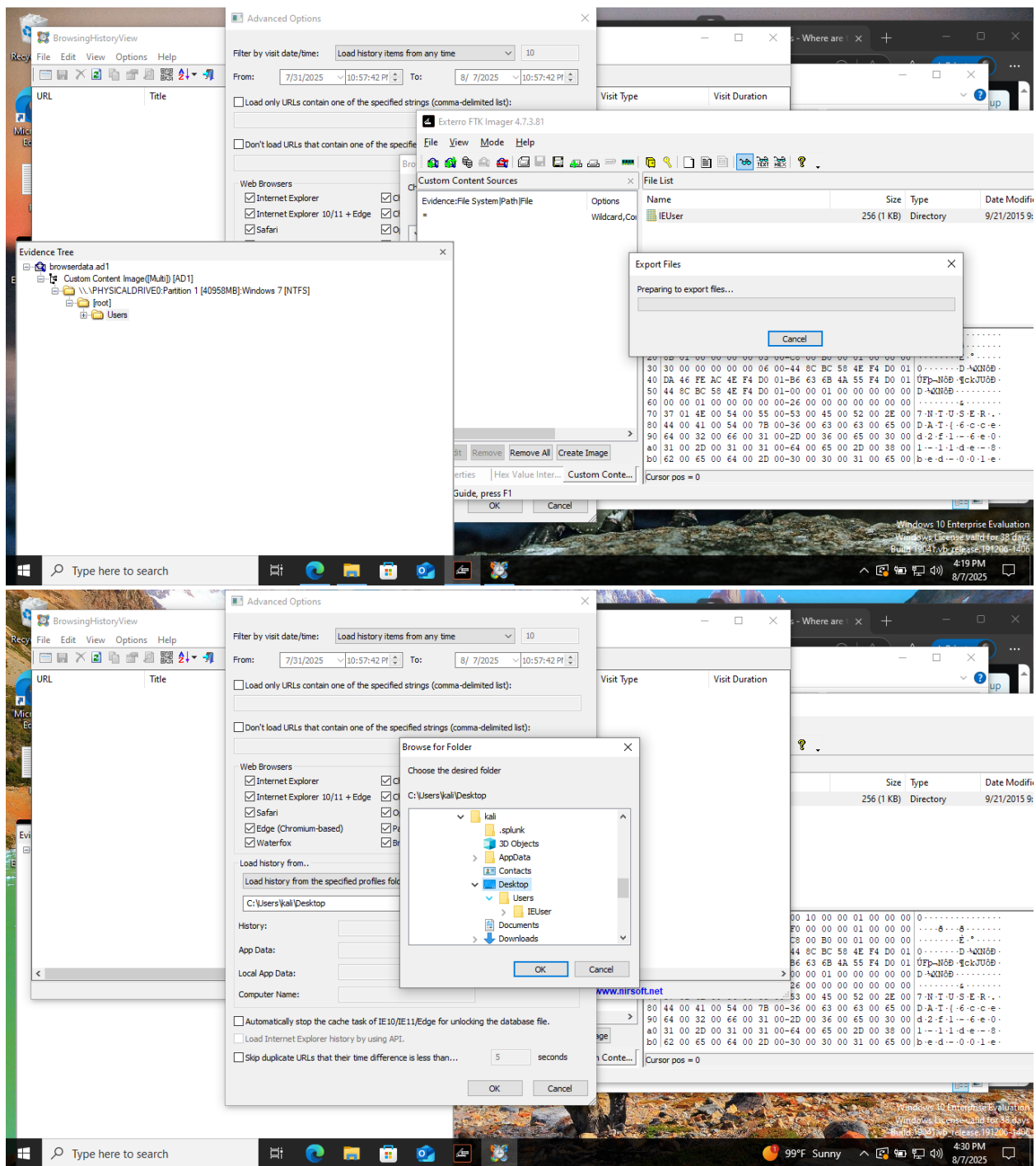
Chrome extensions (including themes) are stored under each profile's `Extensions` directory. Each extension is represented by a folder named with a unique 32-character alphanumeric ID, corresponding to its Chrome Web Store identifier.

To identify which extension represented the installed theme:

1. The `manifest.json` file in each extension folder was reviewed.
2. This file contains metadata such as name, description, version, and permissions.
3. Theme-related extensions are labeled with "theme" in their `manifest.json` properties.

Reviewing these files allowed the identification of the active Chrome theme installed in the environment.





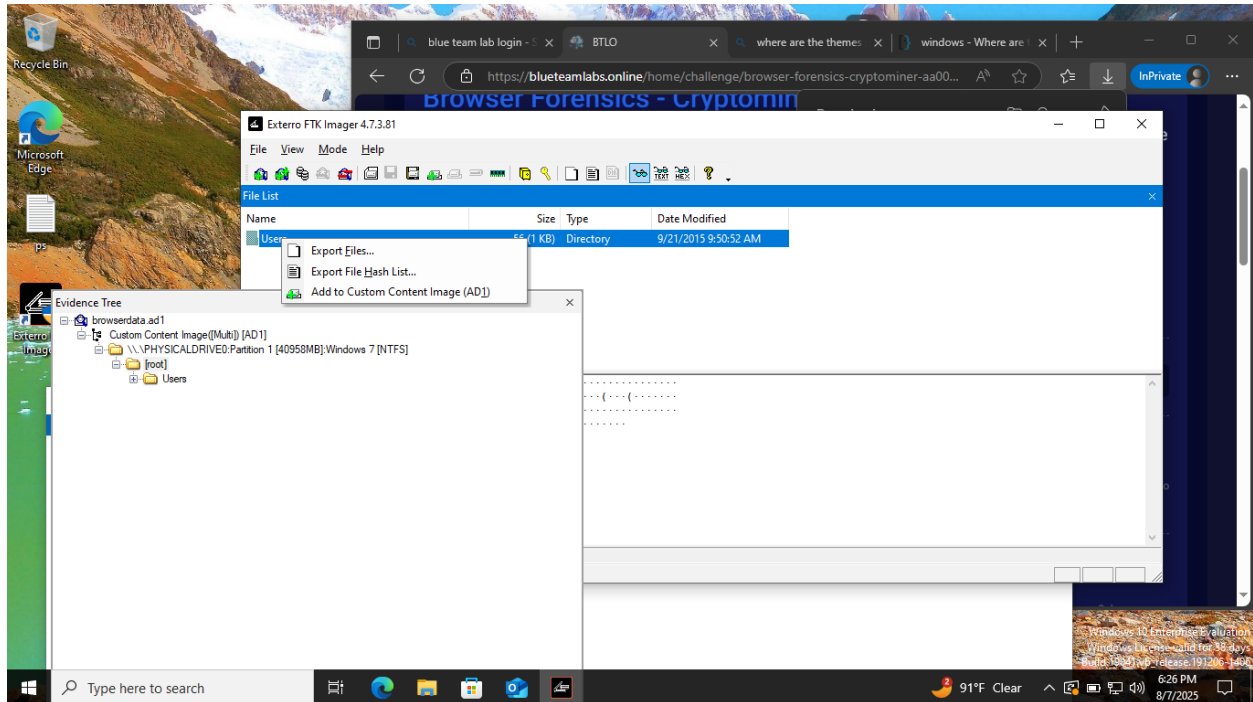
Loading Browser History Data

The initial configuration of Browser History Viewer involved selecting **Load history items from any time** to ensure the analysis captured the full browsing timeline without restrictions. The history source was set to **Load history from the specified profiles folder**. In a normal workflow, this would mean navigating to a local directory such as:

sql

C:\Users\\AppData\Local\Google\Chrome\User Data\

However, in this case, the user profile data was embedded within the AD1 forensic image. To make it accessible to Browser History Viewer, the relevant directory had to be exported from **FTK Imager**.



Browser History Analysis

The **Browser History Viewer** was used to correlate browsing activity with the suspected malicious extension. This tool parses browser history databases (e.g., SQLite .sqlite files for Chrome's History and Web Data stores) and presents them in chronological order.

Configuration steps included:

- Selecting **Load history items from any time** to ensure a complete timeline.
- Pointing the tool to the extracted Chrome profile directories from the AD1 contents.
- Reviewing visited URLs, search queries, and extension activity.

This review focused on:

- Visits to suspicious domains associated with cryptocurrency mining JavaScript scripts.
- Extension update URLs that matched known malicious Chrome extension repositories.
- Evidence of data exfiltration to attacker-controlled servers.

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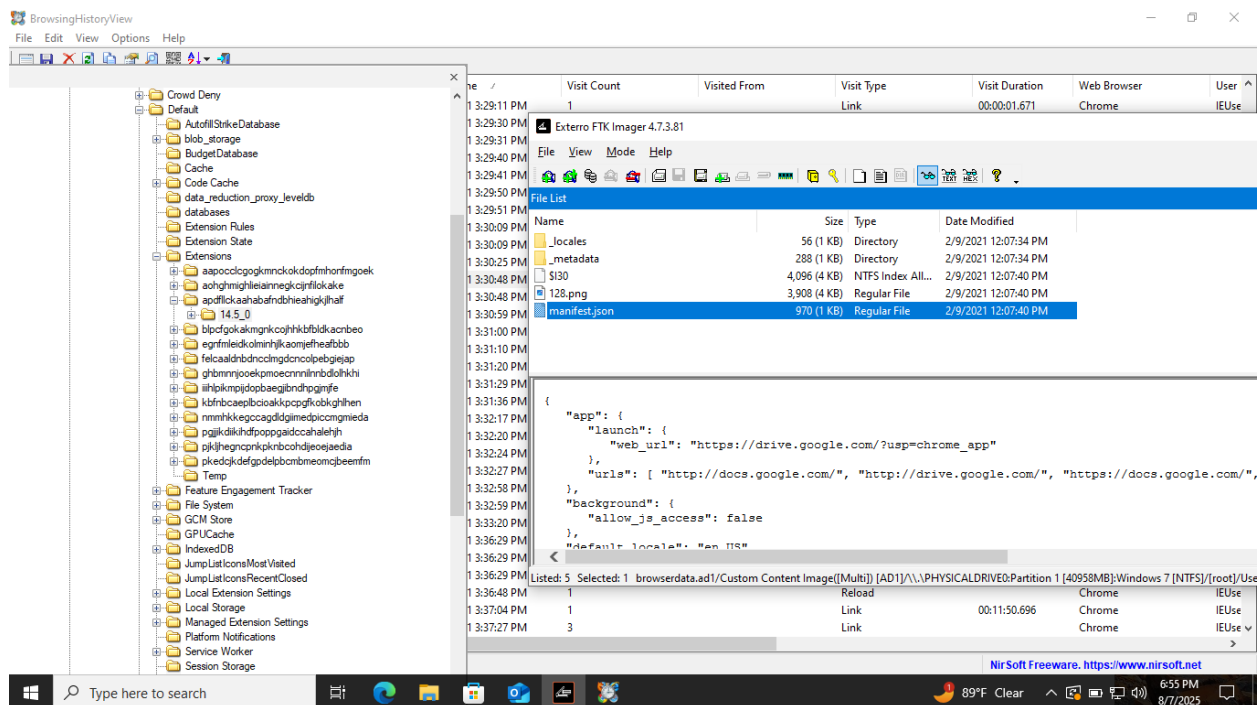
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https://www.theverge...	Google's new Chrome 88 update improves dark mod...	2/9/2021 3:31:20 PM	1		Link	00:00:45.333	Chrome	IEUse
https://chrome.googl...	Earth in Space - Chrome Web Store	2/9/2021 3:31:29 PM	1		Link	00:01:15.251	Chrome	IEUse
https://chrome.googl...	Speedtest by Ookla - Chrome Web Store	2/9/2021 3:31:36 PM	1		Link	00:05:06.625	Chrome	IEUse
https://www.youtube...	Personalize Your Web Browser with Google Chrome T...	2/9/2021 3:32:17 PM	3		Link		Chrome	IEUse
https://www.youtube...	Apply Themes in Google Chrome - YouTube	2/9/2021 3:32:20 PM	3		Link		Chrome	IEUse
https://www.youtube...	How to Create Your Own Chrome Browser Theme - C...	2/9/2021 3:32:24 PM	1		Link	00:05:33.942	Chrome	IEUse
https://www.digitaltre...	The Best Google Chrome Themes Digital Trends	2/9/2021 3:32:27 PM	1		Link	00:00:25.234	Chrome	IEUse
https://www.google.c...	holidays 2021 - Google Search	2/9/2021 3:32:58 PM	2	https://www.google.co...	Form Submit		Chrome	IEUse
https://www.google.c...	holidays 2021 - Google Search	2/9/2021 3:32:59 PM	2	https://www.google.co...	Link		Chrome	IEUse
https://www.youtube...	Top 17 Google Chrome Tips & Tricks - YouTube	2/9/2021 3:33:20 PM	1		Link	00:04:44.177	Chrome	IEUse
https://www.youtube...	Personalize Your Web Browser with Google Chrome T...	2/9/2021 3:36:29 PM	3		Link		Chrome	IEUse
https://www.youtube...	Personalize Your Web Browser with Google Chrome T...	2/9/2021 3:36:29 PM	3		Link		Chrome	IEUse
https://www.youtube...	10 MUST-HAVE chrome extensions for students (for p...	2/9/2021 3:36:29 PM	1		Link	00:02:20.037	Chrome	IEUse
https://chrome.googl...	Speedtest by Ookla - Chrome Web Store	2/9/2021 3:36:48 PM	1		Reload		Chrome	IEUse
https://chrome.googl...	Chrome Web Store - Extensions	2/9/2021 3:37:04 PM	1		Link	00:11:50.696	Chrome	IEUse
https://www.youtube...	Apply Themes in Google Chrome - YouTube	2/9/2021 3:37:27 PM	3		Link		Chrome	IEUse

296 item(s), 1 Selected

NirSoft Freeware. https://www.nirsoft.net

Type here to search

6:53 PM 8/7/2025



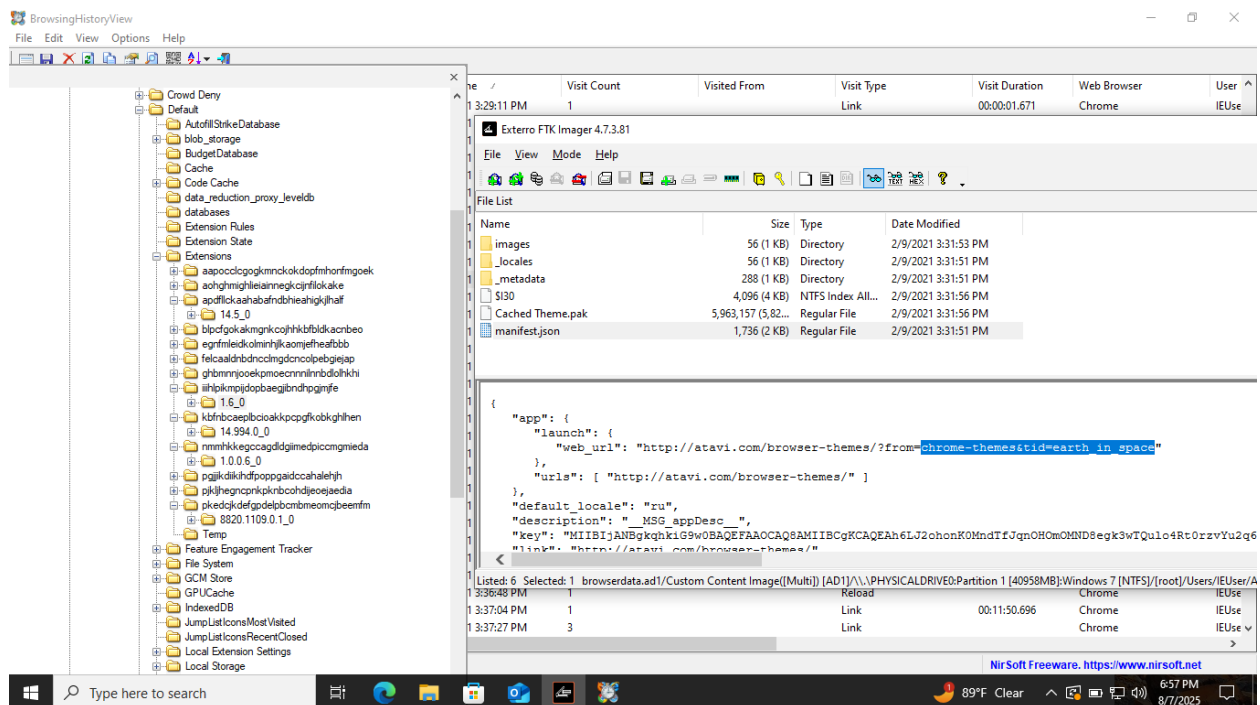
Using History to Identify Installed Theme

One of the lab's questions required determining the installed Chrome theme. The assumption was that the theme installation process would leave traces in browsing history — either via a Chrome Web Store visit or related search queries.

Reviewing the **Title** column in Browser History Viewer revealed a sequence of entries:

- Searches for “Google Chrome themes.”
- Visits to the **Chrome Web Store**.
- A specific theme page titled **Earth in Space**.

This suggested “Earth in Space” was the installed theme. However, verification required cross-referencing with Chrome’s local extension data in the forensic image.



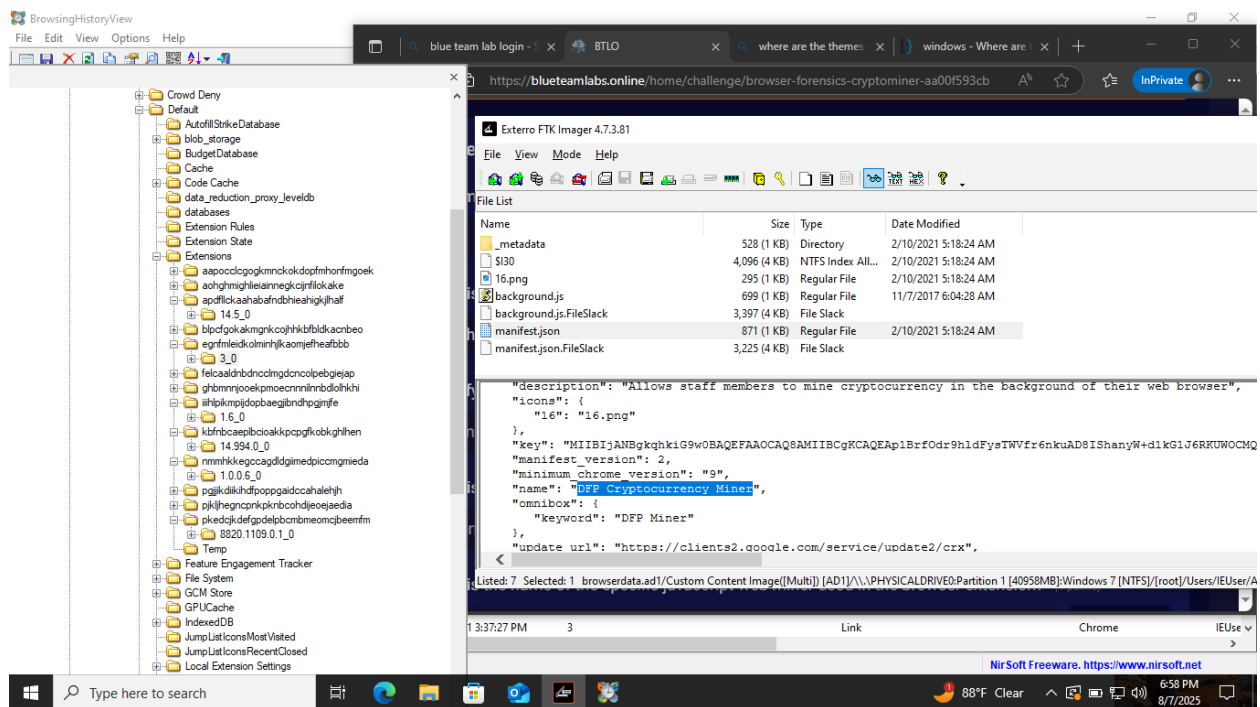
Extension Review in FTK Imager

Chrome extensions are stored under each profile's `Extensions` folder, with each extension identified by a unique alphanumeric directory name. Within each extension folder, the `manifest.json` file describes the extension's name, version, description, and permissions.

Systematic review of each `manifest.json` revealed:

- Standard Google services extensions (Google Drive, YouTube, etc.).
- A notable malicious extension named **DFP Cryptocurrency Miner**.
- Other benign utilities and themes.

Although the **Earth in Space** theme was identified in browsing history, it was also possible to confirm its presence by matching its `manifest.json` entry to the Chrome Web Store ID seen in the browsing history URLs.



The timeline correlation revealed that around the time the user visited the “Earth in Space” theme page (observed at 15:29 UTC in browser history), a corresponding extension folder had been updated. Upon reviewing the `manifest.json` file in this folder, it confirmed the extension’s theme name as “Earth in Space.”

Next, a known malicious extension associated with cryptocurrency mining was identified. The extension ID began with “eg” and ended in “bb,” and the `manifest.json` file listed the name as “**dfp Cryptocurrency Miner.**” This information was documented in the required format:

ExtensionID, ExtensionName.

The description field within the `manifest.json` stated:

“Allow staff members to mine cryptocurrency in the background of their web browser.”

To further investigate, the extension’s files were reviewed. A `background.js` file was identified as potentially containing the web miner’s core logic. Even though no preview data was visible in the forensic tool, the file’s size (699 bytes) confirmed the presence of executable code.

The file was extracted for closer examination, with a security precaution applied—its `.js` extension was renamed to `.malware` to avoid accidental execution. This was done with Windows file extensions visibility enabled, a recommended practice when handling potentially malicious scripts.

The script was opened in Notepad++, revealing a key discovery:

The first line included a `<script>` source pointing to `https://crypto-`

`loot.com/miner.min.js`. Within the code, a variable assignment (`var miner = new`

`CryptoLoot.Anonymous(...)` confirmed the use of **CryptoLoot** as the JavaScript web mining framework.

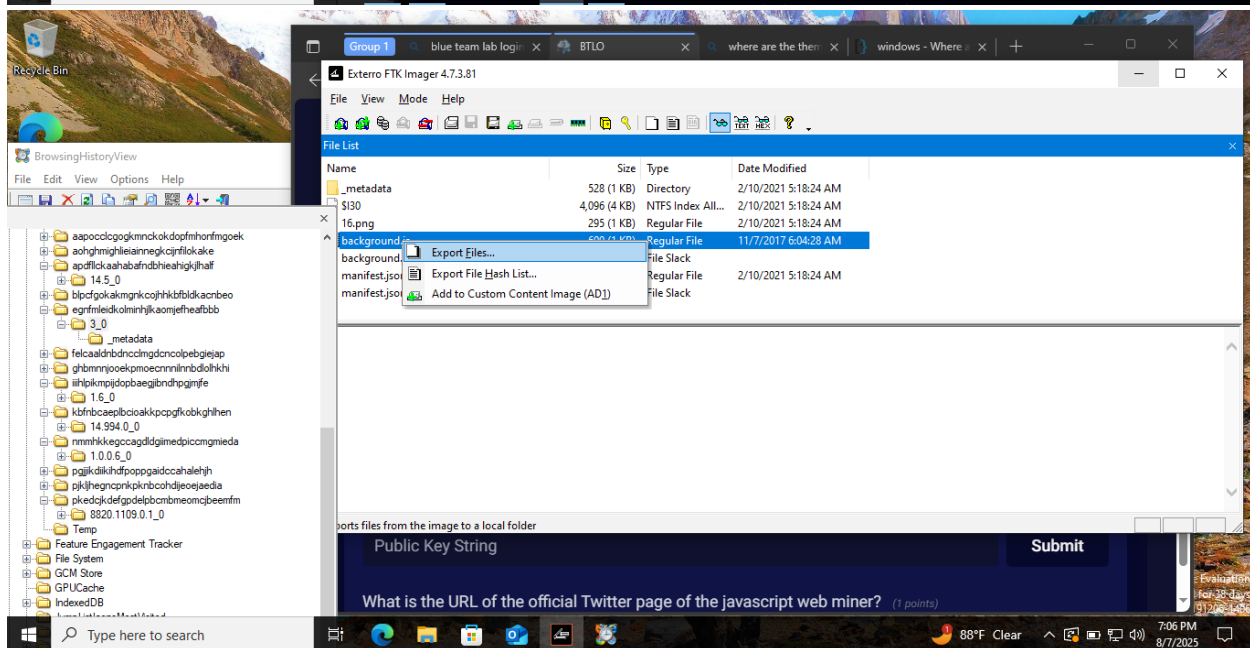
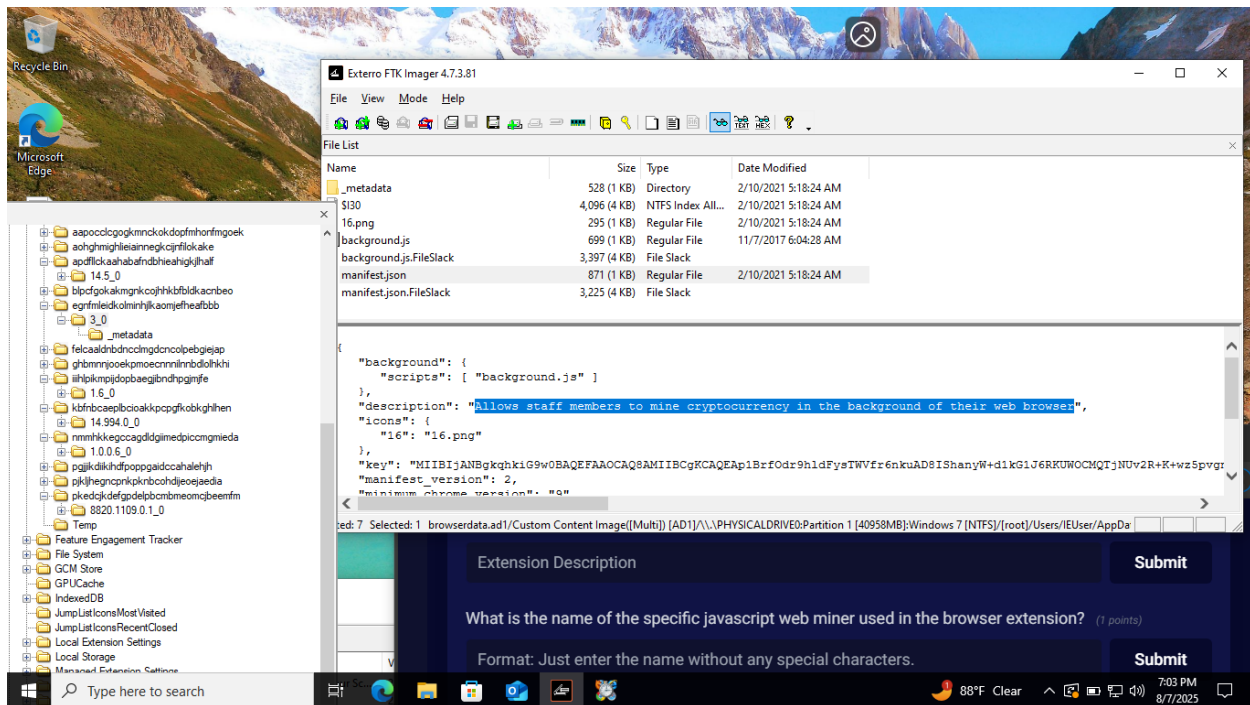
Additional analysis of the code revealed:

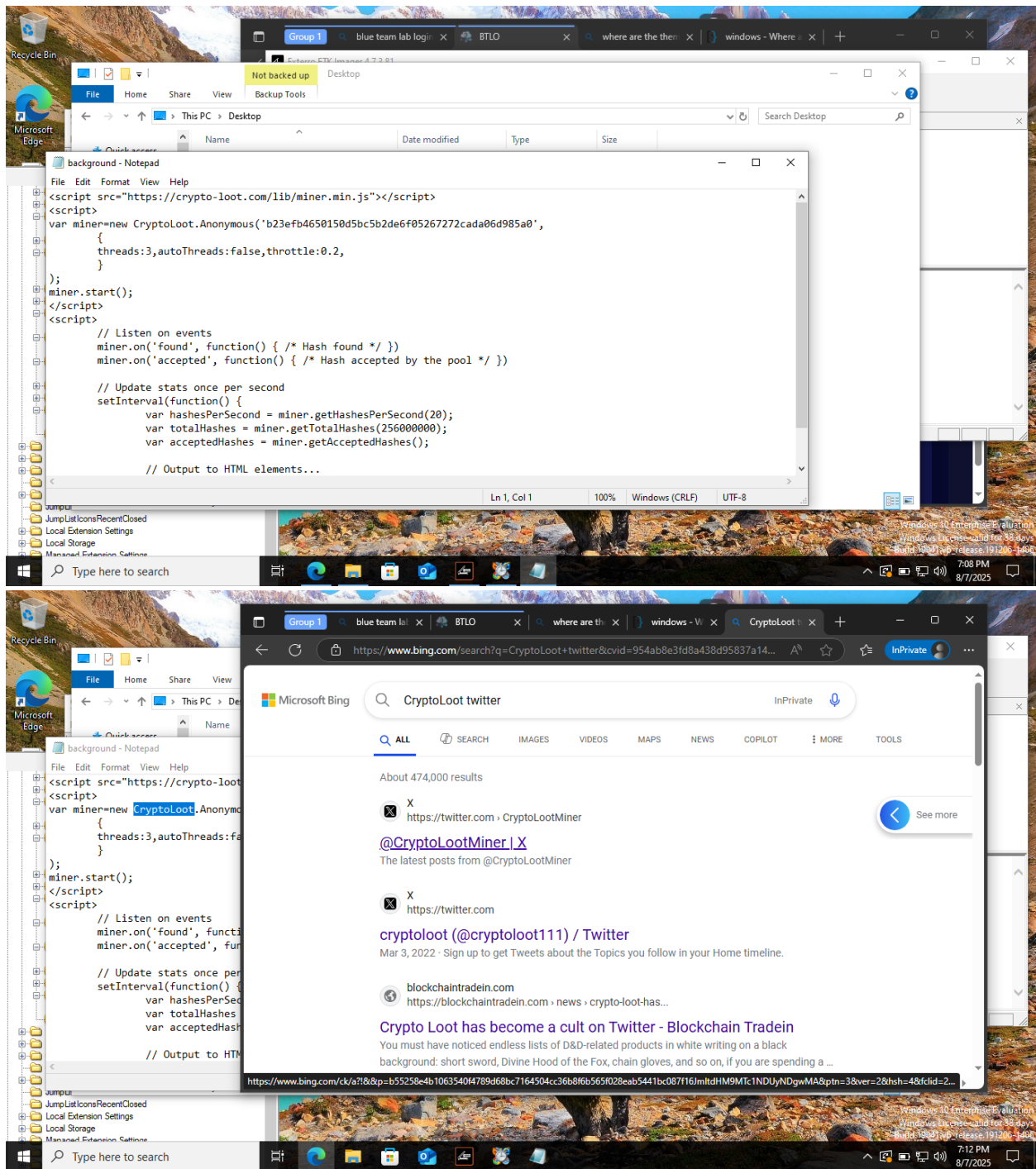
- **Hash rate:** The miner was configured to calculate **20 hashes per second** (`getHashesPerSecond()` returned 20).
- **Public key:** A long alphanumeric string found in the initialization parameters served as the miner's unique public key, linking the activity to its operator.

For attribution purposes, an open-source search was performed on “CryptoLoot Miner” along with “Twitter,” revealing the official profile:

<https://twitter.com/CryptoLootMiner>.

The investigation concluded with the identification of the malicious extension, its description, mining framework, performance metrics, public key, and official associated social media profile.





Conclusion

The investigation confirmed that the suspicious Facebook account activity was linked to a malicious Chrome extension installed on the employee's workstation. The **DFP Cryptocurrency Miner** extension actively used CryptoLoot to mine cryptocurrency, consuming resources and exposing the environment to potential secondary attacks.

Key takeaways:

- **Malicious browser extensions** can be disguised as legitimate productivity tools or cosmetic themes.
- **Logical image acquisition (AD1)** allows targeted evidence collection for efficient forensic review.
- **Timeline correlation** between browsing activity and extension metadata is critical for attribution.
- **Preventive measures** include enforcing browser extension whitelisting, using security scanning tools like CRXcavator, and conducting regular employee awareness training on software installation risks.

Recommendation: Immediately remove the malicious extension, perform a full endpoint scan for residual threats, reset credentials for all potentially compromised accounts, and review browser extension policies to block future unauthorized installations.